Programmatic Modeling of Concurrent Composition and High-Order Observers

Labeled Finite-State Automata

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Problem Setting

Consider a finite-state system G observed by two agents.

- Agent A_1 estimates the current state of G from its observations.
- Agent A_2 has a different observation of the same execution.
- Agent A_2 infers the possible state estimates held by Agent A_1.

Target computation

For an observation β of Agent A_2, compute

$$M_{A_1 \leftarrow A_2}(G, \beta) = \{M_1(G, \ell_1(s)) : s \in L(G),\ \ell_2(s) = \beta\}.$$

LFSA Observer and Concurrent Composition

Labeled finite-state automaton

$$S = (Q, E, \delta, Q_0, \Sigma, \ell), \qquad \ell : E \to \Sigma \cup \{\epsilon\}.$$

The labeling function models what an agent observes.

Observer

$$\delta_{\text{obs}}(q_{0,\text{obs}}, \alpha) = M(S, \alpha) \subseteq Q.$$

Each observer state is one current-state estimate.

Concurrent composition

$$CC(S_1, S_2)$$

It synchronizes transitions with equal observable labels and interleaves unobservable transitions.

Source: Zhang, *Annual Reviews in Control*, 56, 100902, 2023.

Order-2 Observer Pipeline

Stage	Transformation	Program
1	Add Agent A_1 labels to G	`fa_agent_to_lfsa.py`
2	Build A_1's observer	`observer.py`
3	Compose the system and observer	`CC.py`
4	Recover the underlying event system	`lfsa_to_fa.py`
5	Replace labels with Agent A_2's view	`fa_agent_to_lfsa.py`
6	Build A_2's observer	`observer.py`

Programmatic data flow

$$(G, A_1, A_2) \longrightarrow Obs_{A_1} \longrightarrow CC \longrightarrow Obs_{A_1 \leftarrow A_2}(G).$$

Theorem 25 and Example 27

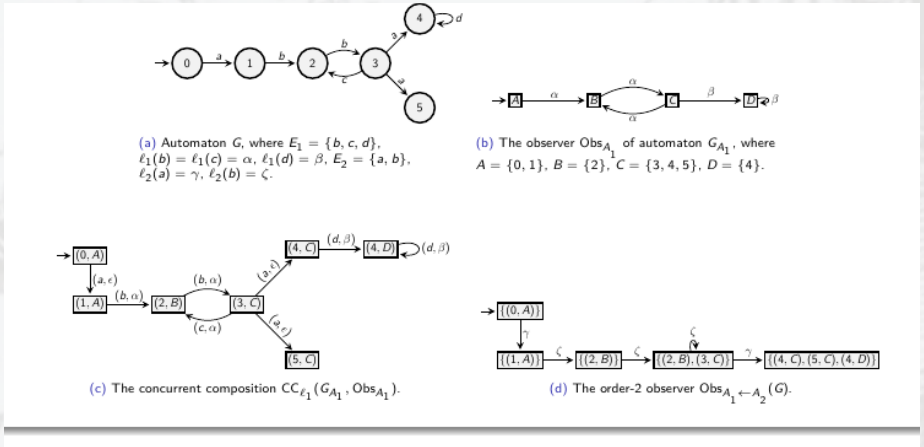
If reading β reaches the order-2 observer state

$$\{(q_1, X_1), \ldots, (q_n, X_n)\},$$

then Theorem 25 gives

$$M_{A_1 \leftarrow A_2}(G, \beta) = \{X_1, \ldots, X_n\}.$$

The second components are precisely the possible estimates of Agent A_1.



The implementation reproduces the published five-state, five-transition order-2 observer.

Reference figure: Zhang et al., *High-Order Observers and High-Order State-Estimation-Based Properties*, Example 27.

Complexity and Summary

Let $n = |Q|$. In the worst case,

$$|Obs_{A_1}| \leq 2^n, \qquad |CC| \leq n2^n, \qquad |Obs_{A_1 \leftarrow A_2}| \leq 2^{n2^n}.$$

Therefore, explicit construction has a double-exponential upper bound in both time and storage, although only reachable states are generated.

Summary

- The programs implement LFSA conversion, observers, and concurrent composition.
- The resulting order-2 observer computes Agent A_2's possible inferences about Agent A_1.
- Example 27 provides an end-to-end validation case.